

with Internet connectivity and mobile data applications via cell phone, there was some data bandwidth in the network that could still cater to this demand.

We need to remind ourselves that the infrastructure here was optimised for voice, but data got mapped onto the voice circuits and transmitted due to the demand. For a long time, Ethernet over SONET/SDH technology was prevalent in terms of how data was backhauled or transmitted over a network that was actually built for efficient voice transmission. However, when the amount of data in the network increased with 3G coming in, and a lot more Internet connectivity and broadband penetration, that was not the most efficient way of transmitting data. That was an expensive alternative and we had to change it.

So what's the more efficient way to transmit data?

Native packets. There are a lot of technologies like Carrier Ethernet that enable you to do that while bringing the best of TDM technologies in terms of protection, fault and monitoring. However, there is still a very large voice network and huge number of 2G subscribers who need to be supported. At some point of time when the data grows, it becomes very expensive to transmit it over a TDM infrastructure. So a separate packet infrastructure has to be built.

What are native packets?

What I mean by native data is that instead of sending it over a voice circuit, we send it over a fibre circuit as an Ethernet packet. Of course, it is necessary to build a lot of other things along with the packet so that issues like fibre cut, data re-routing or fault location in the network are handled. When a network of thousands of network elements is built and spread across the country, it is very important to detect those faults (no traffic, no service) and locate the problem.

How did Tejas Networks contribute to the Indian government's RailTel in setting up a digital TV infrastructure using SDH/DWDM-based transmission systems?

It was built around ELAN technologies that not only provide an efficient transmission of video but also technologies like video multicast that are required for a television transmission infrastructure. ELAN helps to leverage the statistical nature of data traffic and builds data pipes that leverage the 'burstiness' of data traffic (where at some point of time there is very high data traffic and at some other time there is no data traffic at all). Multiple people can share this fixed pipe. So, there is efficient use of bandwidth over a shared infrastructure while transmitting data.

The video is transmitted as packets. So a very high amount of video traffic is transmitted efficiently over the shared packet infrastructure enabled by ELAN. An important benefit of this is the efficient use of bandwidth.

When multicasting of video is done, there is a source from where the TV traffic is derived. The source need not send the same amount of traffic to every user. The data can be sent as common traffic until a certain point and then it can be split. Being able to service so many viewers from a single head-end source but not burning up too much bandwidth is possible by knowing where the traffic should be replicated and transmitted efficiently.

What challenges crop up while implementing communication systems in the field?

Our products are used for multiple applications. One of them is mobile backhaul, for which telecom service providers use our equipment. One of the biggest applications is connecting a base station to the switching centre. The challenges here are setting up a multitude of base stations simultaneously connected to the same switching centre, and ensuring that this set-up performs as a resilient network that is protected against hardware and software failures. For 3G and 4G services, when the user talks to the base station via his handset, a part of the network connects, takes that data from the base station and brings it to all the switching centres. This is known as backhaul.

Could you elaborate further on the backhaul network?

It relates to the transfer of data from any user access point to a central point where it is switched or routed. For instance, we shall consider the backhaul of Internet services. People have broadband connectivity from home, so taking the data from your home to the Internet router is also called backhaul. It is a generic term for taking data from a user access point to a central switching station and redistributing it (wherein it goes to other ends of the network).

What is the most innovative product or technology from Tejas Networks?

An interesting innovation that comes to my mind is the double-bandwidth protected ring (DBPR), which we have patented. It allows doubling of the microwave network capacity. Portion of the network built on microwave is not very high-speed as it is not fibre and goes over the air. If people have a network that drives traffic to packet instead of doing an overhaul of the network by deploying all the IP radios and so on, we provide them a technology called DBPR that lets them double the capacity of data traffic, which could be used intelligently for implementing protection architectures and more. So that is a very interesting technology because without changing the infrastructure or ripping/rebuilding the microwave infrastructure, people can use our products as an add-on to their network and double the capacity of microwave while transmitting data. ●

(The complete interview can be read at electronicsforu.com)